Geometric series – 5.2

Example: $\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$ $\quad a = \frac{1}{2}$, $r = \frac{1}{2}$

Definition: A *geometric series* is one of the form $\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots$

We call a the *initial term* and r the *common ratio*.

Note: $\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=5}^{\infty} ar^{n-5}$

When do these converge?

$\sum_{n=0}^{\infty} r^n$ converges $\Longleftrightarrow$ $\sum_{n=0}^{\infty} ar^n$ converges for $a \neq 0$

enough to say when $\sum_{n=0}^{\infty} r^n$ converges.

If $r=1$, then we're looking at $1+1+1+\dots \to \infty$

which diverges.

If $r \neq 1$, then we have a trick:

take $s_k = 1+r+r^2+\dots+r^k$ and multiply

by $(1-r)$ to get

$$\begin{aligned}(1-r)s_k = 1 &+ r + r^2 + \dots + r^k \\ &- r - r^2 - \dots - r^k - r^{k+1} \\ \hline 1 &- r^{k+1}\end{aligned}$$

$(1-r)s_k = 1-r^{k+1}$ and because $1-r \neq 0$, we get

$$s_k = \frac{1-r^{k+1}}{1-r}.$$

When $|r|<1$, we get $r^{k+1} \to 0$, so the series converges to $\lim_{k\to\infty} \frac{1-r^{k+1}}{1-r} = \frac{1}{1-r}$.

When $|r|>1$, we get $\lim_{k\to\infty} r^{k+1} = \infty$ if $r>1$ or $\lim_{k\to\infty} r^{k+1}$ DNE if $r<-1$, and in both cases, the series diverges.

$$\boxed{\text{For } |r|<1,\ \sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.}$$

Example: What is $\sum_{n=1}^{\infty} 2\left(\frac{-4}{11}\right)^n$?

$$= \frac{-8}{11} + \frac{32}{121} - \cdots$$

Common ratio is $\frac{-4}{11}$

Initial term is $\frac{-8}{11}$

and the series therefore converges since $\left|\frac{-4}{11}\right| < 1$, and equals

$$\frac{a}{1-r} = \frac{\frac{-8}{11}}{1-\left(\frac{-4}{11}\right)} = \frac{\frac{-8}{11}}{\frac{15}{11}} = \frac{-8}{15}.$$

Example: What is $\sum_{n=2}^{\infty} 2^{3n} \cdot 3^{4-2n}$?

$$\sum_{n=2}^{\infty} 2^{3n} \cdot 3^{4-2n} = \sum_{n=0}^{\infty} 2^{3(n+2)} \cdot 3^{4-2(n+2)}$$

$$= \sum_{n=0}^{\infty} 2^{3n+6} \cdot 3^{-2n} = \sum_{n=0}^{\infty} 2^{6} \cdot 2^{3n} \cdot 3^{-2n}$$

$$= \sum_{n=0}^{\infty} 2^{6} \cdot (2^{3})^{n} \cdot (3^{-2})^{n} = \sum_{n=0}^{\infty} 2^{6} \cdot (2^{3} \cdot 3^{-2})^{n}$$

$$= \sum_{n=0}^{\infty} 2^{6} \left(\frac{8}{9}\right)^{n} = \frac{2^{6}}{1-\frac{8}{9}} = \frac{2^{6}}{\frac{1}{9}} = 9 \cdot 2^{6}.$$